Modeling of Option Prices: The Black-Scholes Theorem Framework

Prepared by
Research and Strategy Team
Programming Team

Quantitative Investment Society

University of Calgary

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Black-Scholes Model

Finance Terminology

1.) (C =) Call Option Price:
Also known as the premium is the given right but not an obligation to buy a stock (or other asset) at a specific price (strike price) before a certain date.

2.) (P =) Put Option Price:
Put Option Price gives the right, but not the obligation, to sell a stock at a specific price (strike price) before a certain date. It's the cost of locking in the right to sell the stock later at a guaranteed price, even if the market tanks.

3.) (K =) Strike Price:
The Strike Price is the fixed price at which the holder of an option can buy (in a call option) or sell (in a put option) the underlying asset. This is usually a stock.
> In a call option: If the stock goes above the strike price, you will profit. If it stays below, it is best to let the option expire
> In a put option: If the stock goes below the strike price, you profit. If it stays above, it is best to let the option expire.

4.) (R =) Risk-Free Interest Rate:
The risk-free rate sets the baseline for all other investments. It's the minimum return an investor expects for putting money into any asset. If investors can earn that return with no risk, riskier investment must suffer more.

5.) (T =) Time to Maturity:
Time to maturity is the amount of time left until a financial contract ends. For an option, it is when the right to buy or sell expires.

6.) (N =) Normal distribution:
The normal distribution assumes that financial variables (stock returns, interest rates or portfolio performance) tend to cluster around a mean/average, with symmetrical variation on either side. The probability shows how likely different outcomes are.

7.) (S_0 =) Current Stock Price:
The market price of the underlying asset at the time of valuation

8.) (σ =) Volatility/Sigma:
The standard deviation of the asset's returns, representing price fluctuations

Black-Scholes Model

Thesis:
The Black-Scholes model calculates the fair price of European-style options by using a mathematical formula that considers factors like the current stock price, the option's strike price, time until expiration, interest rates, and market volatility. By assuming that stock prices follow a predictable pattern and that markets are free of friction, the model provides traders and investors with an accurate way to estimate option values and manage financial risk, especially when volatility is stable and the option is held to maturity and/or expiration.

Black-Scholes Model Formula

$$C = SN(d_1) - Ke^{-rt}N(d_2)$$

where:

$$d_1 = \frac{\ln\frac{S}{K} + \left(r + \frac{\sigma_v^2}{2}\right)t}{\sigma_s\sqrt{t}}$$

$$d_2 = d_1 - \sigma_s\sqrt{t}$$

C = Call option price

S = Current stock price

K = Strike price

r = Risk-free interest rate

t = Time to maturity

N = Normal distribution

How do we use it?

1.) Find and research variables that are required for d1 and d2. (Data & analytical team will collect information)

2.) Compute for d1 and d2 (programming team will implement this formula via **python**).

3.) Plug in the answers from d1 and d2 and find **Normal distribution** values using **R code** (pnorm(d1) and pnorm(d2)).

*note pnorm(d1) represents the probability that the strike price is worth investing
e.g.) pnorm(d1) = 0.54, then there's a 54% chance that the **strike price today is worth investing**
*note pnorm(d2) represents the probability that the stock price will <u>exceed</u> the strike price when the option expires.
e.g.) Today's strike price (cost of stock) = $100, pnorm(d2) = 0.62, then there's a 62% chance that the price of the stock will **exceed** above $100

Black-Scholes Model

4.) plug in pnorm(d1), and pnorm(d2) into the C = formula. C will give us the **premium**. if the market price < premium, e.g.) Market price $7, Premium price $8, the option is **undervalued,** potential buy opportunity.
If market price > premium, Market price 9$, Premium 8$, the option is **overvalued**, avoid buying.

5.) If the option is undervalued, our financial team will exercise the option and record it's trajectory as the option expires.

Objective

The object of this project is to apply the Black-Scholes Model in estimating options, specifically European style options using real financial data - yahoo, investopedia (for stock and option parameters) , bankofcanada (for Canadian risk-free interest rates) treasurygov (for US risk-free yield curves) This includes calculating d1, and d2 through the model's formula, and finding the normal distributions for each variable. We then calculate using the C= formula to determine the premium. Based on the premium being undervalued or overvalued, investment transactions will be executed on a virtual $100K CAD simulation environment and the transactions will be recorded till expiration to prove the Black-Scholes theorem.

Code

Black-Scholes Model

```
import math


def norm_cdf(x):
   return 0.5 * (1 + math.erf(x / math.sqrt(2)))

def d1formula(S, K, r, sd, t):
   lnPart = math.log(S / K)
   upper = lnPart + (r + 0.5 * (sd ** 2)) * t
   lower = sd * math.sqrt(t)
   return upper / lower

def d2formula(d1, sd, t):
   return d1 - (sd * math.sqrt(t))

def cformula(S, K, r, t, sd):
   d1 = d1formula(S, K, r, sd, t)
   d2 = d2formula(d1, sd, t)
   Nd1 = norm_cdf(d1)
   Nd2 = norm_cdf(d2)
   return S * Nd1 - K * math.exp(-r * t) * Nd2

price = cformula(S=0, K=0, r=0, t=0, sd=0)
print(f"Option Price: ", price)
```